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THE INFLUENCE OF SULPHIDE ON THE DISTRIBUTION OF HIGHER PLANTS IN SALT MARSHES

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SUMMARY

(1) Soluble sulphide in surface (0–5 cm) salt marsh sediments was detectable only on the lower marsh, salt pans and creek beds. On the lower-marsh only *Salicornia europaea* amongst the vascular plant species present was rooted in sulphide-containing sediments.

(2) No significant correlation was observed between soluble sulphide concentration and redox potential in soil samples from the lower-marsh.

(3) When eight salt marshes from around the coast of Britain were compared, six had a detectable sulphide concentration in the lower-marsh sediments. Divisive information analysis of the vegetation data from these sites indicated that in all cases the most significant association was between *Salicornia europaea* and otherwise bare ground. In two marshes where no soluble sulphide could be measured, *S. europaea* was associated with other plant species rather than bare ground.

(4) Monthly observations of plant cover and sulphide concentration in sixteen permanent quadrats on the lower marsh revealed a significant positive correlation between the cover of *Salicornia europaea* and soil sulphide. In contrast, *Puccinellia maritima* showed a significant negative correlation with sulphide concentration.

(5) In liquid media, the growth of *Atriplex patula*, *Festuca rubra* and *Puccinellia maritima*, was significantly inhibited by sulphide whereas there was no marked effect on that of *Salicornia europaea*.

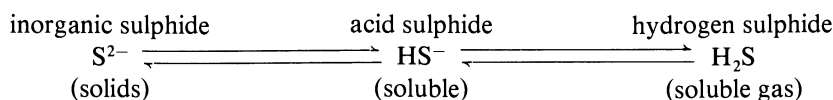
(6) The results suggest that *S. europaea* is relatively tolerant of sulphide and is able to establish on areas of the lower marsh from which other species are excluded by the presence of sulphide.

INTRODUCTION

There is very little published work on the toxic effects of dissolved sulphide on plants. Hollis (1967) studied the so-called ‘soil diseases’ of rice. He observed, that under waterlogged conditions, sulphide production may be associated with disorders such as ‘Akiochi’ a wilting disease which results in reduced yields. Mendelshev (1973) investigated the cause of death of ornamental *Acacia* trees which had been planted in waterlogged soils and manured heavily with seaweed. He attributed the death of the trees to the high concentration of sulphide which occurred under these conditions.

Sulphide is produced in those anaerobic environments where there is a supply of sulphate ions and organic matter adequate for the sulphate-reducing bacteria of the genus *Desulphovibrio* Beijerinck (Pankhurst 1971). These organisms utilize sulphate as a terminal electron acceptor for oxidative phosphorylation. The product, sulphide, is released into the soil where much of it may be taken up by ions of the transition metals, such as iron, copper and manganese, resulting in the formation of insoluble inorganic sulphides. However,

sulphide in soils exists in a dynamic equilibrium, which is partly determined by the activity of hydrogen ions:



Thus, under field conditions, some sulphide remains in solution both as the acid sulphide ion (HS^-) and as hydrogen sulphide which, although a gas, is highly soluble. In areas of high sulphate-reducing activity, hydrogen sulphide is gradually lost from solution to the atmosphere. Thus the presence of appreciable soluble sulphide in soils will depend upon continued microbial sulphide production. The cycling of sulphur in freshwater and marine sediments has been described by Postgate (1979) and Nedwell (1982).

One environment in which sulphide production is known to occur is salt marsh (Nedwell & Abram 1978). A high concentration of dissolved sulphide in salt marsh sediments has been correlated with reduced growth of *Spartina townsendii* agg. (Goodman & Williams 1961) and *S. alterniflora* Lois (De Laune, Buresh & Patrick 1979; Mendelssohn, McKee & Patrick 1981; King *et al.* 1982). However, none of these studies has provided any direct experimental evidence that sulphide is toxic to plants over the range of concentrations found in the field. In particular, it remains to be established whether the reduced growth of some plants in salt marsh sediments is caused by sulphide *per se* or is attributable to a low soil-redox potential.

This investigation was designed to assess the importance of sulphide as an ecological factor in salt marshes. If the presence of sulphide in soils has any significant ecological consequences for vascular plants, some demonstrable effect on their distribution should be shown. Between-species differences in sulphide resistance might also influence the frequency with which species occur together. Thus the pattern of association between species might vary with the concentration of dissolved sulphide in the sediment.

THE STUDY AREAS

Most of the field work was carried out on the salt marsh at Canvey Island Point (National Grid reference TQ 837831) on the Essex coast of the river Thames estuary. The marsh, which is about 130 ha in area, has formed within the protection of a shell and shingle spit. A characteristic feature of the marsh is an extensive middle marsh, dominated by *Halimione portulacoides* (L.) Aellen. The upper-marsh area is much reduced due to sea wall construction. Some regions of the lower marsh have steep erosion cliffs up to 1 m in height. The dominant species in the lower-marsh (i.e. that zone below the seaward limit of *H. portulacoides*) which formed the main study area are *Salicornia europaea* L. and *Puccinellia maritima* Huds. although there is local invasion by *Spartina anglica* Hubbard.

Other salt marshes studied were: (i) Spurn Point (TA 405116), located on the north side of the Humber estuary on the leeward side of a 5-km long sand bar; (ii) Blakeney Point (TG 027457) on the North Norfolk coast; the marsh has formed in the lee of a shingle bank; (iii) Colne Point (TM 107123), which is an estuarine marsh formed at the mouth of the river Colne (near Colchester) in Essex; (iv) Two Tree Island (TQ 832848), located in the Thames estuary on the Essex coast and formed in the lee of a barrier island; (v) West Itchenor (SU 796016), in the Chichester channel of the Chichester harbour basin; (vi) Axmouth (SY 254902), a narrow small marsh, formed in the sheltered estuary of the river

Axe; and (vii) Conwy (SH 796774) on the sheltered east bank of the river Conwy, Clwd, N. Wales.

METHODS

Field measurements

To ascertain whether sulphide production occurs throughout salt marshes or is restricted to certain areas within them, measurements were made of soil sulphide concentration (see below) at 200 random locations on Canvey Island Point salt marsh. The relatively large number of samples ensured that topographical features such as creek beds, creek banks and salt pans were adequately represented.

Soil sulphide concentration was then compared with plant distribution. Observations were made on the lower-marsh areas at each of the seven sites listed previously. The presence of individual plant species was recorded in June–August in between 250 and 375 completely randomized (25×25 cm) quadrats at each site; the actual number varying between sites. Bare ground was recorded where appreciable areas of uncolonized sediment occurred between colonies or groups of individuals. The floristic data obtained from this study were analysed using the divisive information analysis technique originally described by Lance & Williams (1968). Additionally, the soil sulphide concentration was measured in the centre of each quadrat to enable comparisons to be made with the observed plant distribution.

To examine the possibility that distribution and soil sulphide concentration might be indirectly related via the effects of a low soil redox potential, both sulphide concentration and redox potential were measured, as closely together as possible, at eighty random points on the lower marsh at Canvey Island Point. At each location any species rooted in the soil were recorded.

Studies were also carried out on sixteen permanent quadrats (25×25 cm). These were located close to the seaward limit of the Canvey Island Point salt marsh and arranged in a 4×4 block so that approximately half of the quadrats lay on an area where the soil was initially found to contain an appreciable sulphide concentration, and the other half on soil lacking detectable soluble sulphide. Within these quadrats, at monthly intervals, plant cover was determined subjectively and soil sulphide concentration was measured at the centre of the quadrat. The data from these permanent quadrats were used *inter alia* to calculate, for each sampling date, correlation coefficients relating species cover to soil sulphide concentration.

Soil analysis

Free sulphide concentration

Sulphide was measured *in situ* with an ion-selective electrode, calibrated on each occasion that it was used. As the output of the electrode is influenced by the hydrogen-ion concentration of the soil solution, simultaneous pH measurements were taken using a toughened-glass combination-electrode and these values were used to adjust the sulphide readings to neutrality. Corrections for pH were small as all measurements fell within the range pH 5.0–6.5.

Where appropriate both sulphide concentration and pH were measured as closely as possible to the centre of the quadrat. The electrodes were slowly inserted to a depth of 5 cm and the highest value obtained was recorded. The sulphide electrode had a lower

detection limit of 5×10^{-6} M. The results obtained by this method were found to be very similar to those obtained using other analytical techniques (Beaton & Burns 1968).

Redox potential

Measurements of redox potential were made *in situ* using a platinum electrode with a silver-silver chloride reference half-cell. Since hydrogen-ion concentration also influences redox potential, the soil pH measurements were used to calculate appropriate adjustments to the observed values. All measurements were made in the top 5 cm of soil as described for soil sulphide determination. A correction of +220 mV has been applied to the recorded results in order that they may be expressed as Eh.

Laboratory experiments

Laboratory experiments were carried out to measure any effects of sulphide on the growth of four common salt marsh species. The species chosen were two perennial grasses, *Festuca rubra* L. and *Puccinellia maritima* and two annual members of the Chenopodiaceae, *Atriplex patula* L. and *Salicornia europaea*. Both *Puccinellia maritima* and *Salicornia europaea* are amongst the earliest vascular plant colonizers of the lower-marsh sediments, whereas *Festuca rubra* and *Atriplex patula* are restricted to the upper-marsh and strand line respectively.

Culture Solution

The composition of the culture solution is shown in Table 1. Nitrogen was supplied as 1 mM calcium nitrate. Iron was provided in a chelated form (FeEDTA) to prevent precipitation in solutions containing sulphide. Sulphide was added to the culture solution,

TABLE 1. Composition of the basic culture solution used for experiments on salt-marsh plant sulphide tolerance.

Element	Concentration	
	mm	$\mu\text{g l}^{-1}$
Nitrogen	2	28
Potassium	1	39
Phosphorus	0.5	15
Sulphur	1.5	49
Calcium	1	20
Magnesium	0.5	12
Iron		1
Manganese		0.4
Copper		0.02
Zinc		0.1
Boron		0.5
Molybdenum		0.1
Cobalt		0.006

All solutions contained 2% sodium chloride.

where appropriate, to give a final concentration of 100 μM sodium sulphide. The amount of sodium, added as sodium sulphide, was small (0.1%) compared with the total sodium content of the medium. The pH value was adjusted to 6.0 by the addition of dilute HCl or NaOH.

Solutions were changed twice-weekly. The sulphide concentration (where appropriate)

and pH were corrected daily as necessary. Solutions were regularly checked for the presence of ammonium using the colorimetric method of McCullough (1967). No detectable ammonium ion occurred in any of the sample aliquots.

Since the addition of sulphide to the growth solution lowers the redox potential, the sulphide solution differs from the standard culture solution not only in the presence of sulphide but also in the redox potential. Attempts to separate these effects by the addition of reducing agents to the control solutions were unsuccessful because such substances were either themselves very toxic or did not bring about a sufficient depression of the redox potential.

Consequently the plant growth studies alone cannot be used to discriminate between the effects of sulphide *per se* and the influence of a reduced redox potential. It will be shown later however, that this distinction can be drawn on the basis of field observations.

Environmental conditions and culture vessels

A controlled environmental cabinet was used to provide constant irradiance (42 W m^{-2}), temperature (23°C) and humidity (68%).

A major technical difficulty associated with the study of the effects of sulphide on plant growth is the rapid rate of loss of sulphide from solutions as hydrogen sulphide gas. It was therefore necessary to design a growth vessel which could be sealed as far as possible from the atmosphere, whilst allowing for the topping-up of solution lost by evapotranspiration from the plants ($<10 \text{ ml}$ per day).

The vessel design is shown in Fig. 1. It consisted of a 500 ml flask with three necks. The central neck was covered with a rubber membrane (0.5 mm thick) in which holes were punched to retain and seal around the seedlings. One side-arm was fitted with a vaccine bottle cap to facilitate sampling with a hypodermic needle and syringe. The other side-arm was fitted with a separating funnel to allow for the frequent topping-up of the solution.

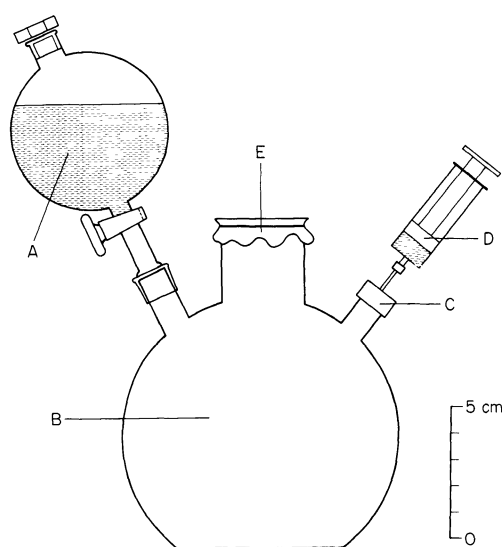


FIG. 1. Diagram of plant culture vessel used in studies on plant sulphide tolerance. Key: (A), separating funnel containing growth solution; (B), black-painted three-necked flask; (C), vaccine bottle cap; (D), hypodermic syringe; (E), rubber membrane.

Experimental procedure

Seeds of all four species were collected from the salt marsh when ripe, dried at room temperature and stored at 4 °C until required for use. The seeds were placed on moist filter paper in Petri dishes and allowed to germinate in a controlled environmental cabinet. When large enough to handle (21–28 days), the seedlings were transferred to 1-litre polypropylene boxes containing standard culture solution and grown for a further two weeks before being used for experiments.

Ten seedlings of each species were taken for initial dry-weight determination. A further four seedlings were placed in each anaerobic growth vessel (see above) and three replicates of each species and treatment (standard culture solution or culture solution containing sulphide) were randomized in a 6 × 4 block. One seedling from each vessel was harvested for dry weight determination after 2 weeks and thereafter at weekly intervals.

RESULTS

Field investigations

Sulphide was detectable at Canvey Island Point salt marsh only in the lower marsh (mean 80 µM), creek beds (mean 400 µM) and salt pans (mean 300 µM). As the creek beds and salt pans are, for the most part, devoid of higher plant vegetation, the remaining investigations were confined to the lower-marsh area. Here a detectable sulphide concentration was found at only about 25% of the sampling locations (range 5–400 µM). The highest concentration was found within 1–2 cm of the soil surface and was invariably associated with a discrete black horizon normally not more than 2–3 mm thick. In some samples these bands were repeated several times within the top 5 cm of sediment. Microscopic examination revealed a conspicuous presence of microalgal remains in this material including several species of both blue-green and green algae.

Only one species, *Salicornia europaea*, was associated with sulphide-containing soils at any of the salt marshes studied. Although this species was often found to be growing in sediments without detectable soluble sulphide, soils supporting *S. europaea* frequently contained appreciable sulphide, with a maximum observed concentration of 400 µM. Moreover, whereas other species commonly occurred in the same quadrat, in those quadrats with measurable sulphide, *S. europaea* was the only higher plant species present.

The divisive information analysis of the vegetation data from Canvey Island Point salt marsh shows a total information content of 1600 units (Fig. 2). The first division of species separates *S. europaea* and bare ground from the remainder, producing a fall in the information content of 800 units.

Soluble sulphide was detectable at five of the other marshes studied: Colne Point; Blakeney Point; Spurn Point; West Itchenor; and Two Tree Island (Table 2). For these sites, as for Canvey Island, the greatest fall in information content was achieved by dividing *S. europaea* and bare ground from the other lower marsh species.

A quite different pattern emerges from the data for the Conwy and Axmouth marshes where the first division groups *S. europaea* with *Aster tripolium* L., *Halimione portulacoides* and *Suaeda maritima* (L.) Dumort, rather than bare ground. Notably, at neither of these sites was any soluble sulphide detected.

Comparison of the sulphide concentration and redox potential at eighty random points on the lower marsh at Canvey Island Point shows that the two factors are not significantly correlated (Fig. 3). Many of the most reduced soil samples contained no measurable sulphide, whereas sulphide was sometimes present in relatively oxidized sediment.

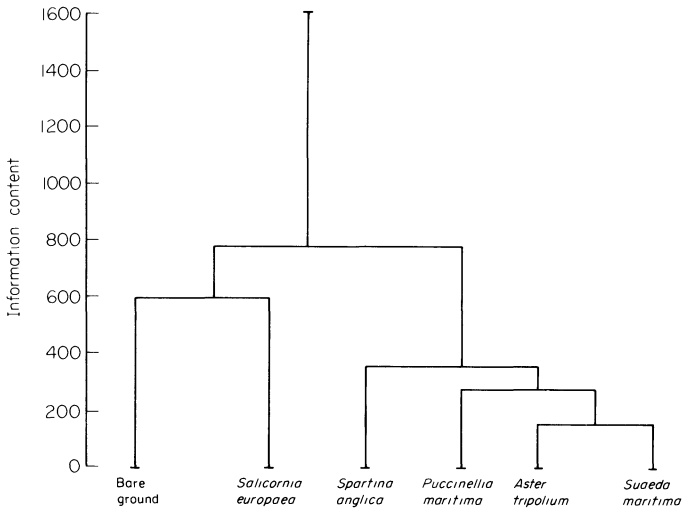


FIG. 2. Classification of 'species' based on divisive information analysis of three hundred and seventy-five 25 × 25 cm quadrats on the lower part of Canvey Island Point salt marsh.

TABLE 2. Divisive information analysis of lower-salt marsh vegetation in relation to maximum free sulphide concentration measured in the soil of eight British salt marshes.

Site	Most significant association	Total information content	Information fall	Maximum tree sulphide concentration (μm)
Canvey Island Point (Essex)	<i>Salicornia europaea</i> bare ground	1600	800	400
Two Tree Island (Essex)	<i>Salicornia europaea</i> bare ground	1100	500	380
Colne Point (Essex)	<i>Salicornia europaea</i> bare ground	1050	450	250
Blakeney Point (Norfolk)	<i>Salicornia europaea</i> bare ground	1275	475	300
Spurn Point (Humberside)	<i>Salicornia europaea</i> bare ground	1200	450	500
West Itchenor (Sussex)	<i>Salicornia europaea</i> bare ground	1100	450	210
Axmouth (Devon)	<i>Aster tripolium</i> <i>Halimione portulacoides</i> <i>Salicornia europaea</i>	800	400	not detectable
Conwy (Clwd)	<i>Suaeda maritima</i> <i>Puccinellia maritima</i> <i>Spartina anglica</i> bare ground	1100	400	not detectable

Although the lowest redox potential measurements were frequently associated with *S. europaea* soils, they were not restricted to this species. Both *Atriplex tripolium* and *Spartina anglica* grew in soils with an Eh of less than 0 mV.

Data obtained from the sixteen permanent quadrats support the apparent association of *S. europaea* with sulphide-rich soils (Table 3). In each month the cover of *S. europaea* was

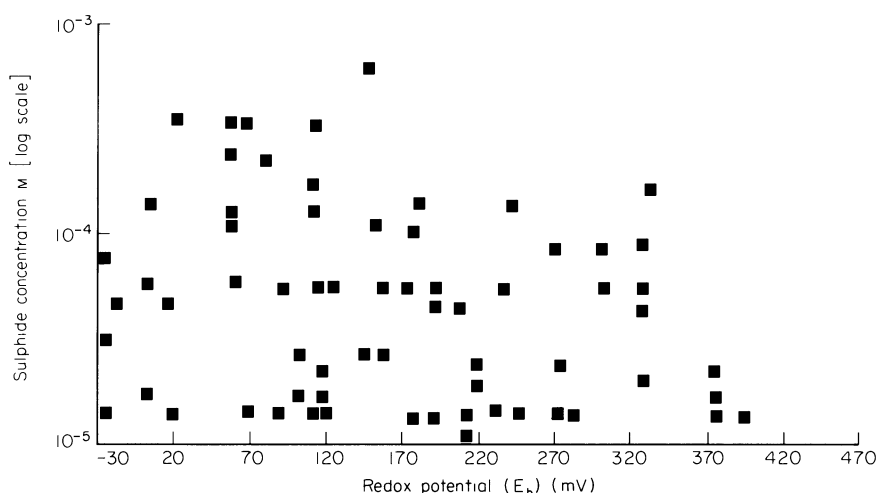


FIG. 3. Scatter diagram showing soil sulphide concentration and redox potential in soil samples from the lower-marsh at Canvey Island Point ($r = -0.208$; $n = 80$, $P < 0.05$).

TABLE 3. Correlation coefficients (r) showing the relationship of plant cover (%) of *Salicornia europaea* and *Puccinellia maritima* to sulphide concentration in British salt marshes. There were 25 samples on each occasion.

Species	Month			
	June	July	September	October
<i>Salicornia europaea</i>	0.41	0.55	0.63	0.50
<i>Puccinellia maritima</i>	-0.57	-0.55	-0.92	-0.82

All correlations were significant at $P < 0.05$.

significantly positively correlated with sulphide concentration. By contrast, the cover of *Puccinellia maritima* showed a significant negative correlation with soil sulphide concentration and this species had greatest cover in those quadrats from which sulphide was absent throughout the summer (Fig. 4). In these areas only occasional etiolated plants of *S. europaea* were seen. Where sulphide did not reach detectable concentrations until July the performance of *P. maritima* was reduced and this was associated with a greater cover of *S. europaea*. Continuing this trend, in quadrats where sulphide could be measured throughout the summer, *S. europaea* was the dominant species and *P. maritima* was either absent or sparse.

Effects of sulphide on plant growth

The influence upon plant growth of adding sulphide to the culture solution is shown in Fig. 5. *Salicornia europaea* made a similar dry weight increase in both the control and sulphide treatments but the growth of the other three species was markedly reduced by sulphide. Control plants of all species showed good root and shoot development with the extensive formation of secondary and tertiary roots. However, amongst the sulphide-treated plants only in *S. europaea* was root development similar to that of the control. In each of the other three species the shoots were chlorotic and the root system was stunted and bore few lateral branches.

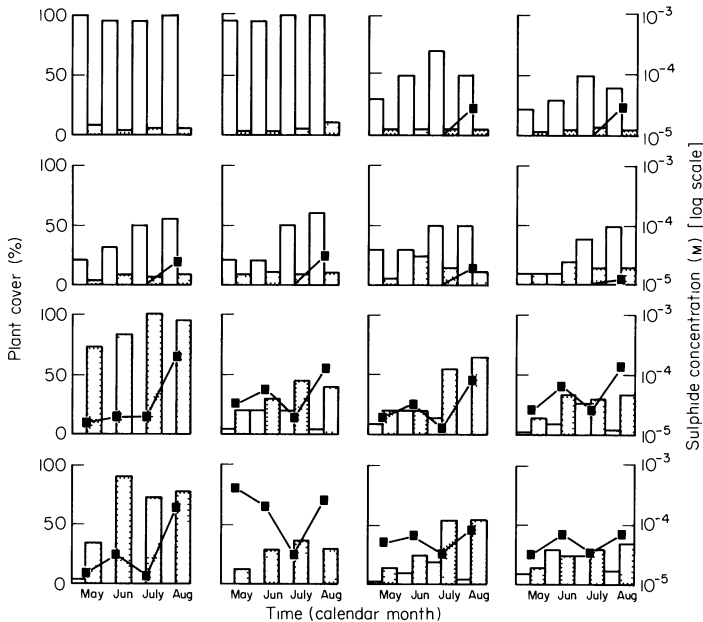


FIG. 4. Monthly variation in cover of *Salicornia europaea* and *Puccinellia maritima* together with sulphide concentration, in sixteen permanent quadrats (25×25 cm) in a 4×4 contiguous array on the lower marsh at Canvey Island Point. Symbols: ($\square$), *Puccinellia maritima*; ($\blacksquare$), *Salicornia europaea*; ($\blacksquare$). Soil sulphide concentration.

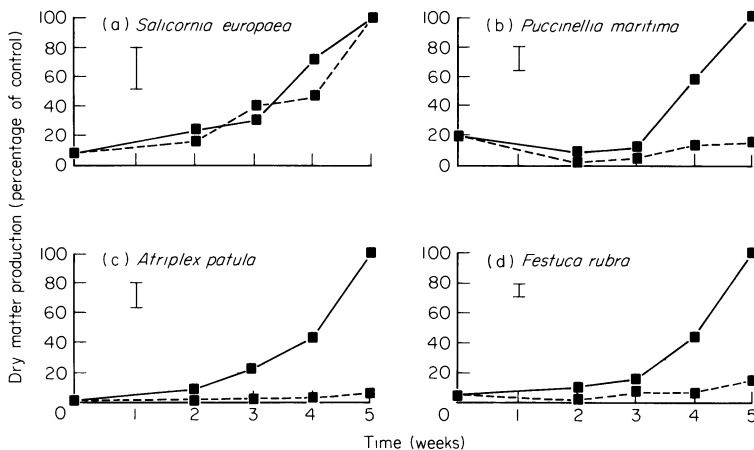


FIG. 5. The effects of $100 \mu\text{M}$ sulphide on the growth of (a) *Salicornia europaea*, (b) *Puccinellia maritima*, (c) *Atriplex patula*, and (d) *Festuca rubra*. Growth of control plants grown without added sulphide is shown by solid lines. Growth of sulphide-treated plants is shown by the broken lines. Vertical bars show the least significant range ($P = 0.05$).

DISCUSSION

The distribution of plants in waterlogged soils is influenced by a number of environmental factors, amongst which the presence of potential plant toxins may be of considerable importance. In the anaerobic sediments which characterize the seaward parts of salt

marshes, microbial activity leads to the production of sulphide, reduced iron (Fe^{2+}) and manganese (Mn^{2+}) along with various hydrocarbons, and a reduced redox potential. It is probably because of this complexity that so few attempts have been made to study the influence of these factors individually.

This investigation has shown, however, that microsite variation in soil redox potential in the lower marsh is not directly correlated with sulphide concentration. The lowest calculated Eh values for Canvey Island Point are higher (minimum -30 mV) than those sometimes recorded for anaerobic salt marsh sediments (see e.g. Armstrong 1982). Moreover, microbial sulphate reduction does not begin until the Eh (pH 7) falls to about -150 mV or lower (Ponnamperuma 1972). It is possible, however, that sulphide production in the surface 5 cm of sediment is restricted to one or more narrow and discrete bands associated with organic material. The method of measuring redox used in this investigation, by direct insertion of the electrode into the sediment, has the advantage that oxidation of samples during collection and storage is avoided. Unfortunately, the electrode will inevitably carry with it some oxidised surface material resulting in a disturbance of the soil profile and a consequent underestimation of the minimum Eh value. The lack of correlation between the dissolved sulphide concentration and redox potential may be attributed to sulphide mobility within the sediment. Sulphide in solution will diffuse towards the soil surface and, presumably, also move laterally by flow of ground water. Thus the precise locations at which dissolved sulphide can be detected do not necessarily represent sites of sulphate reduction.

The interstitial water of salt-marsh sediments may contain high concentrations of water-soluble Mn^{2+} and Fe^{2+} . However the reduction of manganese and iron occurs at higher Eh values (c. $+200$ mV and $+120$ mV, respectively) than sulphate reduction (Ponnamperuma 1972). Accordingly, the distribution of these elements is quite different to that of sulphide (Singer & Havill 1984; D. C. Havill unpublished). Although anaerobic sediments undoubtedly contain a wide variety of other reduced substances including ammonium and hydrocarbons, any influence which these may have upon plant growth can be excluded by performing suitable plant culture experiments using defined media. It is suggested, therefore, that where the distribution of plant species is closely correlated with sulphide concentration, it is possible to attribute these effects to the influence of sulphide *per se*.

King *et al.* (1982) suggested that sulphide toxicity inhibited the growth of *Spartina alterniflora* only on the higher levels of the marsh where lateral soil water movement and the iron concentration are below those found on the lower-marsh. Moreover, in their investigation sulphide concentration was found to increase with depth down to 10 cm below the soil surface. In contrast, at the Canvey Island Point salt marsh soluble sulphide was present at a concentration greater than $5\text{ }\mu\text{M}$ only in creek beds, salt pans and parts of the lower marsh. Within the lower-marsh area soluble sulphide concentration showed considerable spatial variation, presumably reflecting an underlying pattern in the microtopography and physiochemical status of the sediment. Where soluble sulphide occurred, the concentration exhibited seasonal changes, reaching a maximum in mid to late summer. The restricted occurrence of sulphide and its association with discrete soil horizons close to the soil surface again contradicts the findings of King *et al.* (1982). One explanation may be that the activity of sulphate reducing bacteria in lower-salt marsh sediments is limited by the availability of usable carbon sources. Accordingly, sulphide production may depend on the release of suitable substrates during the decomposition of algal mats.

The toxicity to plants of soluble sulphide is clearly indicated by the reduced growth of *Atriplex patula*, *Festuca rubra* and *Puccinellia maritima* when grown in culture solutions containing sulphide. In contrast, the dry matter production of *Salicornia europaea* plants was not significantly affected by sulphide nor were any visual symptoms of toxicity observed. Thus it would appear that the sulphide-tolerance of *S. europaea* is greater than that of at least some other salt marsh species.

The ecological importance of soluble sulphide in this environment is indicated by the virtual absence of species other than *S. europaea* from sulphide-containing soils. One obvious implication of these results is that *S. europaea* is capable of establishing on sediments with a relatively high sulphide concentration from which other species, with a lower tolerance, are excluded. The data obtained from the permanent quadrats at Canvey Island Point support this hypothesis. The significant positive correlation between sulphide concentration and plant cover for *S. europaea* is in marked contrast to the inverse relationship between sulphide and *P. maritima*. These results do not imply that *S. europaea* has any requirement for sulphide, indeed *S. europaea* plants often establish in soils where no sulphide can be detected. It would appear however that, as in the culture experiments, the presence of sulphide in sediment reduces the growth of *P. maritima*, producing areas of bare ground in which seedlings of the annual *S. europaea* may become established. According to this argument, it is the distribution of perennial species such as *P. maritima* which is restricted by sulphide, enabling *S. europaea* to exploit the resultant areas of bare ground.

This association between *S. europaea* and bare ground is clearly shown by the information analyses of the floristic data from salt marshes where soluble sulphide was detectable. In each case the greatest fall in information content was achieved by separating *S. europaea* and bare ground from the remaining species. Furthermore, it is noticeable that in those western salt-marshes where no soluble sulphide could be measured, this association was not detected.

Although not recorded in this survey, it should be appreciated that other species of the genus *Salicornia*, notably *S. dolichostachya* Moss and *S. perennis* Miller also establish on the lower-marsh. We are currently investigating the distribution of these and other *Salicornia* species in relation to the soil sulphide concentration.

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